

MagicCode Super-Resolution

Software Specification

1. Document Overview

1.1 Purpose

This document formally defines the technical specifications, functional capabilities, system requirements, and performance benchmarks of the MagicCode Super-Resolution Software (hereinafter referred to as "MagicCode"). It serves as an authoritative reference for the software's development, testing, quality assurance, deployment, and user documentation, ensuring that all stakeholders have a consistent and clear understanding of the product's design objectives and functional boundaries.

1.2 Scope

MagicCode is a specialized image super-resolution application designed to enhance the resolution of digital images by integer multiples while maintaining or optimizing image detail. The scope of this specification encompasses the following core aspects: supported operating systems and hardware platforms, instruction set compatibility, super-resolution functional parameters, input resolution constraints, multi-threading optimization mechanisms, and specific performance indicators on designated devices.

1.3 Definitions, Acronyms and Abbreviations

Term/Acronym	Definition
Super-Resolution (SR)	A digital image processing technology that increases the spatial resolution of an image, generating high-resolution (HR) images from low-resolution (LR) inputs
x86_64	A 64-bit extension of the x86 instruction set architecture (ISA), widely used in desktop and laptop

	computers
ARM64	A 64-bit version of the ARM ISA, commonly adopted in mobile devices, tablets and some laptops
SIMD	Single Instruction, Multiple Data, a parallel processing technology that enables a single instruction to operate on multiple data elements simultaneously
NEON	An advanced SIMD architecture extension for the ARM ISA, enhancing multimedia and signal processing performance
AVX2	Advanced Vector Extensions 2, an extension of the x86 instruction set that expands SIMD capabilities for high-performance computing
4K	A video resolution standard typically referring to 3840×2160 pixels
fps	Frames Per Second, a unit measuring the number of image frames displayed or processed per second

1.4 Document Status

Version	Date	Author	Status	Change Description
V1.0	2025-12-01	Technical Documentatio	Initial Release	First formal version of the

		n Team		specification
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2. System Requirements

2.1 Operating System Compatibility

MagicCode supports multiple mainstream operating systems across desktop and mobile platforms, with specific version requirements as follows:

- Windows: Supports Windows 10 (64-bit) and above
- Android: Supports Android 8.0 and above
- iOS: Supports iOS 13.0 and above
- iPadOS: Supports iPadOS 13.0 and above
- macOS: Supports macOS 12.0 (Monterey) and above

2.2 Hardware Requirements

2.2.1 CPU Architecture Support

The software is compatible with the following 64-bit CPU architectures to ensure broad hardware adaptability:

- x86_64: Applicable to mainstream Intel and AMD desktop/laptop CPUs
- ARM64: Applicable to Apple Silicon (e.g., M-series chips) and mainstream ARM-based mobile CPUs

2.2.2 SIMD Instruction Set Support

To optimize image processing performance, MagicCode leverages advanced SIMD instruction sets for parallel computing, with mandatory support for the following instruction sets based on CPU architecture:

- NEON: Mandatory for ARM64 architecture CPUs, enabling efficient multimedia data processing on mobile and Apple Silicon devices
- AVX2: Mandatory for x86_64 architecture CPUs, providing enhanced vector processing capabilities for desktop platforms

3. Functional Specifications

3.1 Super-Resolution Scaling Ratios

MagicCode supports three fixed integer super-resolution scaling ratios, allowing users to select the appropriate ratio based on their quality and efficiency needs:

- 2x scaling: Doubles the width and height of the input image, resulting in a 4x total pixel count increase
- 3x scaling: Triples the width and height of the input image, resulting in a 9x total pixel count increase
- 4x scaling: Quadruples the width and height of the input image, resulting in a 16x total pixel count increase

Note: The software automatically adapts the internal algorithm parameters based on the selected scaling ratio to balance output quality and processing speed.

3.2 Super-Resolution Modes

Two super-resolution processing modes are provided to meet different application scenarios (e.g., real-time processing vs. offline enhancement). Specific performance indicators on the iPhone 14 Pro (a representative ARM64 device) are defined as follows:

Mode	Characteristics	Performance on iPhone 14 Pro (4K Input)
Veryfast	Prioritizes processing speed, suitable for real-time scenarios such as video playback	Supports 4K@60fps super-resolution processing
Fast	Balances processing speed and output quality, suitable for offline image/video enhancement	Supports 4K@30fps super-resolution processing

3.3 Input Resolution Constraints

To ensure processing stability and efficiency, MagicCode defines clear constraints on the input image resolution (width × height):

- Maximum input resolution: 7680×7680 pixels (8K UHD standard)
- Minimum input resolution: 32×32 pixels

If the input image resolution exceeds the maximum limit or is lower than the minimum limit, the software will return an error prompt and refuse to process. Users need to adjust the input resolution to the valid range first.

4. Performance Optimization Specifications

4.1 Multi-Threading Processing Mechanism

MagicCode adopts a dynamic multi-threading strategy to improve processing efficiency for large-size images. The activation conditions and constraints of multi-threading are as follows:

- Activation condition: The number of input image rows (height) is ≥ 256 . For images with fewer than 256 rows, single-threaded processing is used to avoid thread scheduling overhead.
- Maximum thread count: 8 threads. The software automatically adjusts the actual number of active threads based on the current CPU core count and system resource usage, but will not exceed 8 threads to prevent resource contention.

On multi-core CPU platforms, this mechanism can significantly reduce the total processing time for high-resolution images, especially when processing 4K and 8K content.

5. Conformance Criteria

All functional and performance indicators specified in this document shall be verified through formal testing. The software shall be deemed conformant only if it meets 100% of the mandatory requirements outlined herein. Any deviations from the specifications must be approved by the product management and technical steering committees, with corresponding revisions to this document.

6. References

1. ARM Architecture Reference Manual: NEON Extension
2. Intel 64 and IA-32 Architectures Software Developer's Manual: AVX2 Instructions
3. ISO/IEC 29100: Information technology — Security techniques — Privacy framework